

Accessing List Elements

List elements can be individually accessed using the index while a contiguous group of elements can be fetched using a slice operation which is identical to string slicing.

```
>>> l[1]
2.2
>>> l[0:2] # result includes start index but not end index
[1, 2.2] # elements corresponding to index 0 and 1 are returned
```

We can also access the list elements by iterating over the list and collecting elements at each index.

```
>>> for i in l:
...     print(i)
...
1
2.2
None
abc
```

List Manipulation

There are numerous ways in which way we can add, delete or update the elements of a `list` in python.

- The simplest and easiest way of updating an element is by using the element index and assigning a new value to the corresponding slice of the list.

```
>>> l = [1, 2, 3, 4, 5]
>>> l[0] = 9
>>> print(l)
[9, 2, 3, 4, 5]
```

- In order to add an element or a sequence of elements to the list, we can use the functions `append` or `extend` respectively. The `append` function takes the element as the parameter while the `extend` function only accepts an iterable as a parameter.